7-1-1: Examining Cloud Concepts

At the end of this episode, I will be able to:

1. Identify and explain cloud concepts, given a scenario.

Learner Objective: *Identify and explain concept concepts, given a scenario.*

Description: In this episode, the learner will examine the concepts of cloud technologies. We will explore cloud deployment models, service types such as SaaS, PaaS, IaaS and more.

- What is Software as a Service?
    - A cloud computing model that delivers applications over the Internet, allowing users to access software applications from any device with an Internet connection without the need for local installation.
- What is Platform as a Service?
    - An environment for developers to create, deploy, and manage application code, in the cloud. This eliminates the need for managing the runtime environment, operating systems, and hardware.
- What is Infrastructure as a Service?
    - Offers virtualized computing resources that can be accessed via the Internet, enabling users to scale and manage hardware, storage, and network capabilities online without physical hardware investment.
- What is the Shared Responsibility Model in the cloud important?
    - Understanding that the security and management tasks are divided between Cloud service providers (CSPs) and customers.
    - Identifying these responsibilities vary by service model.
- What are examples of these responsibilities?
    - IaaS (Infrastructure as a Service)
        - CSP responsibilities
            - Physical servers, storage, and networking.
        - Customer responsibilities
            - Operating systems, applications, data, and access management.
    - PaaS (Platform as a Service)
        - CSP responsibilities
            - Physical servers, storage, networking, runtime environment, operating systems.
        - Customer responsibilities
            - Applications, data, and access to the environment.
    - SaaS (Software as a Service)
        - CSP responsibilities
            - Managing infrastructure (physical servers, storage, middleware, networking), applications, and security of the platform.
        - Customer Responsibilities
            - Managing user access and protecting user data.
- What is elasticity in cloud computing?
    - This refers to the ability to dynamically scale computing resources up or down based on demand.
    - This feature allows for efficient resource utilization, ensuring that applications can handle varying workloads without manual intervention.
- What is scalability in the cloud?
    - The capability to expand or reduce resources and services according to the changing demands of applications
    - This enables systems to accommodate growth seamlessly, ensuring consistent performance and user experience as workload requirements evolve.

    - Note - the focus on scalability is the ability of the system to grow, long-term

- What is on-demand access in the cloud?
    - This allows users to automatically provision computing resources as needed without requiring human interaction with each service provider.
    - This chracteristic enables immediate access and control over resources according to the users' requirements.

- What is broad network access in the cloud?
    - Ensures that services are available over the network and can be accessed through standard mechanisms by a wide range of devices, including:
        - Mobile phones
        - Tablets
        - Laptops
        - Workstations
    - This characteristic allows users to
        - Access applications and data from anywhere
        - Provides mobility and convenience

- Additional Resources:
    - NIST = National Institute of Standards and Technology
    - NIST SP 800-145 (defines the terms for Cloud): https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-145.pdf